*Worked-Out Numerical Exercise for Solving the IS–LM Model*

*By Instructors : Macroeconomics-I, 2025-26*

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## Learning Objectives

By the end of this exercise, you should be able to:

- Derive the IS and LM curves algebraically from behavioral equations
- Find short-run and long-run equilibrium values
- Understand the adjustment mechanism from short-run to long-run equilibrium
- Derive the aggregate demand (AD) curve from IS-LM framework
- Verify results using the AS-AD model

## Problem Setup

This worked-out numerical exercise demonstrates the complete algebra needed for solving the IS–LM model. We'll work through each step systematically, explaining the economic intuition behind the mathematics.

The IS-LM model, developed by John Hicks (1937), provides a framework for understanding the interaction between the goods market and money market.

**The Economy:** Consider an economy described by the following behavioral equations:

$$C^d = 300 + 0.75(Y - T) - 300r \quad \text{(Consumption function)} \quad (1)$$

$$T = 100 + 0.2Y \quad \text{(Tax function)} \quad (2)$$

$$I^d = 200 - 200r \quad \text{(Investment function)} \quad (3)$$

$$L = 0.5Y - 500i \quad \text{(Money demand function)} \quad (4)$$

**Parameters and Initial Conditions:**

- Potential output: $\bar{Y} = 2500$
- Government spending: $G = 600$
- Money supply: $M = 133,200$
- Expected inflation: $\pi^e = 0.05$
- Short-run price level: $P_{sr} = 120$ (fixed)

**Objective:** Find the short-run and long-run equilibrium values of $Y$, $P$, $r$, $C$, $I$, and $i$.

## Step 1: Deriving the IS Curve

The IS curve represents combinations of output ($Y$) and real interest rate ($r$) that ensure equilibrium in the goods market. The equilibrium condition is:

$$Y = C^d + I^d + G$$

The IS curve slopes downward because higher interest rates reduce both consumption and investment, leading to lower equilibrium output.

### Substituting the Tax Function

First, we substitute the tax function into the consumption equation:

$$\begin{aligned} C^d &= 300 + 0.75[Y - (100 + 0.2Y)] - 300r \\ &= 300 + 0.75Y - 75 - 0.15Y - 300r \\ &= 225 + 0.6Y - 300r \end{aligned}$$

### Finding Goods Market Equilibrium

Now we can write the equilibrium condition:

$$\begin{aligned} Y &= C^d + I^d + G \\ Y &= [225 + 0.6Y - 300r] + [200 - 200r] + 600 \\ Y &= 1025 + 0.6Y - 500r \end{aligned}$$

Solving for $r$ in terms of $Y$:

$$\begin{aligned} Y - 0.6Y &= 1025 - 500r \\ 0.4Y &= 1025 - 500r \\ 500r &= 1025 - 0.4Y \\ r &= 2.05 - 0.0008Y \end{aligned}$$

**IS Curve:** $r = 2.05 - 0.0008Y$
The negative coefficient on $Y$ confirms the downward slope. For every 100-unit increase in output, the real interest rate must fall by 0.08 percentage points to maintain goods market equilibrium.

## Step 2: Deriving the LM Curve

The LM curve represents combinations of $Y$ and $r$ that ensure equilibrium in the money market.

The Fisher equation links nominal and real interest rates: $i = r + \pi^e$

### General Form of the LM Curve

The money market equilibrium condition is:

$$\frac{M}{P} = L$$

Since $L = 0.5Y - 500i$ and $i = r + \pi^e$, we have:

$$\begin{aligned} L &= 0.5Y - 500(r + 0.05) \\ &= 0.5Y - 500r - 25 \end{aligned}$$

Setting money supply equal to money demand:

$$\begin{aligned} \frac{133,200}{P} &= 0.5Y - 500r - 25 \\ 500r &= 0.5Y - 25 - \frac{133,200}{P} \\ r &= 0.001Y - 0.05 - \frac{266.4}{P} \end{aligned}$$

**General LM Curve:** $r = 0.001Y - 0.05 - \frac{266.4}{P}$
This equation shows how the LM curve position depends on the price level. Higher prices reduce real money supply, shifting the LM curve upward.

### Short-Run LM Curve

In the short run, prices are fixed at $P_{sr} = 120$:

$$\begin{aligned} r &= 0.001Y - 0.05 - \frac{266.4}{120} \\ &= 0.001Y - 0.05 - 2.22 \\ &= 0.001Y - 2.27 \end{aligned}$$

**Short-Run LM Curve:** $r = 0.001Y - 2.27$
The LM curve slopes upward with a gradient of 0.001, meaning a 100-unit increase in output requires a 0.1 percentage point rise in the interest rate.

The positive slope of the LM curve reflects that higher income increases money demand, requiring higher interest rates to restore money market equilibrium.

## Step 3: Short-Run Equilibrium

The short-run equilibrium occurs at the intersection of the IS and LM curves.

### Finding Equilibrium Output and Interest Rate

Setting IS = LM:

$$\begin{aligned} 2.05 - 0.0008Y &= 0.001Y - 2.27 \\ 2.05 + 2.27 &= 0.001Y + 0.0008Y \\ 4.32 &= 0.0018Y \\ Y &= 2400 \end{aligned}$$

Substituting back into either curve to find $r$:

$$\begin{aligned} r &= 2.05 - 0.0008(2400) = 2.05 - 1.92 = 0.13 \\ \text{or } r &= 0.001(2400) - 2.27 = 2.40 - 2.27 = 0.13 \end{aligned}$$

Notice that short-run output (2400) is below potential output (2500), indicating a recessionary gap.

### Other Short-Run Variables

Now we calculate the remaining variables:

**Tax Revenue:**

$$T = 100 + 0.2(2400) = 580$$

**Consumption:**

$$\begin{aligned} C &= 300 + 0.75(2400 - 580) - 300(0.13) \\ &= 300 + 1365 - 39 = 1626 \end{aligned}$$

**Investment:**

$$I = 200 - 200(0.13) = 174$$

**Nominal Interest Rate:**

$$i = r + \pi^e = 0.13 + 0.05 = 0.18$$

**Verification:** Always check that components sum to output:

$$C + I + G = 1626 + 174 + 600 = 2400 = Y \checkmark$$

## Step 4: Long-Run Equilibrium

In the long run, output returns to its potential level, and prices adjust to clear all markets.

The long-run adjustment involves price flexibility, allowing the economy to return to full employment.

### Finding the Long-Run Real Interest Rate

In long-run equilibrium, $Y = \bar{Y} = 2500$. Using the IS curve:

$$\begin{aligned} r &= 2.05 - 0.0008(2500) \\ &= 2.05 - 2.00 \\ &= 0.05 \end{aligned}$$

Note that $r = 0.05 = \pi^e$, illustrating the Fisher effect in long-run equilibrium.

### Long-Run Components

**Tax Revenue:**

$$T = 100 + 0.2(2500) = 600$$

**Consumption:**

$$\begin{aligned} C &= 300 + 0.75(2500 - 600) - 300(0.05) \\ &= 300 + 1425 - 15 = 1710 \end{aligned}$$

**Investment:**

$$I = 200 - 200(0.05) = 190$$

**Nominal Interest Rate:**

$$i = 0.05 + 0.05 = 0.10$$

Compare with short-run values: both consumption and investment are higher in the long run due to lower interest rates.

### Finding the Long-Run Price Level

To find the long-run price level, we use money market equilibrium:

**Money Demand:**

$$\begin{aligned} L &= 0.5(2500) - 500(0.10) \\ &= 1250 - 50 = 1200 \end{aligned}$$

**Money Market Equilibrium:**

$$\begin{aligned} \frac{M}{P} &= L \\ \frac{133,200}{P} &= 1200 \\ P &= 111 \end{aligned}$$

**Key Insight:** Prices fall from 120 to 111 in the adjustment from short run to long run. This price decline increases real money supply, shifting the LM curve rightward until it intersects the IS curve at full employment output.

## Step 5: Deriving the AD Curve

The AD curve shows all combinations of $P$ and $Y$ consistent with simultaneous equilibrium in both goods and money markets.

The AD curve is derived by eliminating the interest rate from the IS-LM system.

Setting IS = LM (general form):

$$
\begin{aligned}
2.05 - 0.0008Y &= 0.001Y - 0.05 - \frac{266.4}{P} \\
2.10 &= 0.0018Y - \frac{266.4}{P} \\
0.0018Y &= 2.10 + \frac{266.4}{P} \\
Y &= 1166\frac{2}{3} + \frac{148,000}{P}
\end{aligned}
$$

**AD Curve:** $Y = 1166\frac{2}{3} + \frac{148,000}{P}$
The negative relationship between $P$ and $Y$ arises from the real balance effect: higher prices reduce real money supply, raising interest rates and reducing output.

## Step 6: Verification Using AS-AD Framework

We can verify our results using the aggregate supply and aggregate demand framework.

### Short-Run Analysis

In the short run:

- SRAS: $P = P_{sr} = 120$ (horizontal line)
- Equilibrium: Substitute into AD curve:

$$
Y = 1166\frac{2}{3} + \frac{148,000}{120} = 1166\frac{2}{3} + 1233\frac{1}{3} = 2400
$$

This confirms our Step 3 result.

### Long-Run Analysis

In the long run:

- LRAS: $Y = \bar{Y} = 2500$ (vertical line)
- Equilibrium: Substitute into AD curve:

$$
\begin{aligned}
2500 &= 1166\frac{2}{3} + \frac{148,000}{P} \\
1333\frac{1}{3} &= \frac{148,000}{P} \\
P &= 111
\end{aligned}
$$

This confirms our Step 4 result.

The AS-AD framework provides an alternative visualization of the same equilibrium conditions derived from IS-LM analysis.

## Summary and Economic Interpretation

**Equilibrium Comparison**

| Variable | Short Run | Long Run |
|---|---|---|
| Output (Y) | 2400 | 2500 |
| Price Level (P) | 120 | 111 |
| Real Interest Rate (r) | 0.13 | 0.05 |
| Nominal Interest Rate (i) | 0.18 | 0.10 |
| Consumption (C) | 1626 | 1710 |
| Investment (I) | 174 | 190 |

### The Adjustment Process

The movement from short-run to long-run equilibrium involves several key adjustments:

1. **Output increases to potential:** The economy moves from a recessionary gap (Y = 2400) to full employment (Y = 2500).
2. **Prices fall:** The price level declines from 120 to 111, reflecting deflationary pressure when output is below potential.
3. **Interest rates decline:** Both real and nominal interest rates fall, stimulating consumption and investment.
4. **Composition of output changes:** Lower interest rates cause both consumption and investment to increase, while government spending remains constant.

This adjustment mechanism assumes price flexibility in the long run—a key assumption of classical economics.

### Economic Intuition

The short-run equilibrium with output below potential creates downward pressure on prices. As prices fall, real money supply increases, shifting the LM curve rightward. This process continues until the economy reaches full employment, where all markets clear simultaneously.

The long-run equilibrium exhibits several important properties:

- The real interest rate equals expected inflation (Fisher effect)
- Output equals potential (classical dichotomy)
- All markets clear (general equilibrium)

**Important Note:** This analysis assumes:
- Fixed expected inflation ($\pi^e = 0.05$)
- No supply shocks
- Price flexibility in the long run
- Stable money demand function

Relaxing these assumptions would alter the adjustment path and equilibrium values.

## Key Takeaways

This exercise demonstrates several crucial macroeconomic principles:

1. **Systematic approach:** Complex models can be solved step-by-step using algebraic methods.
2. **Market interdependence:** Goods and money markets interact through the interest rate channel.
3. **Short-run vs. long-run:** Price stickiness creates temporary deviations from potential output.
4. **Automatic adjustment:** Price flexibility eventually restores full employment.
5. **Policy implications:** Understanding these mechanisms is essential for monetary and fiscal policy analysis.

This framework forms the foundation for more advanced macroeconomic models, including New Keynesian DSGE models.